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(54) **CROSSING BRIDGE AND METHOD**

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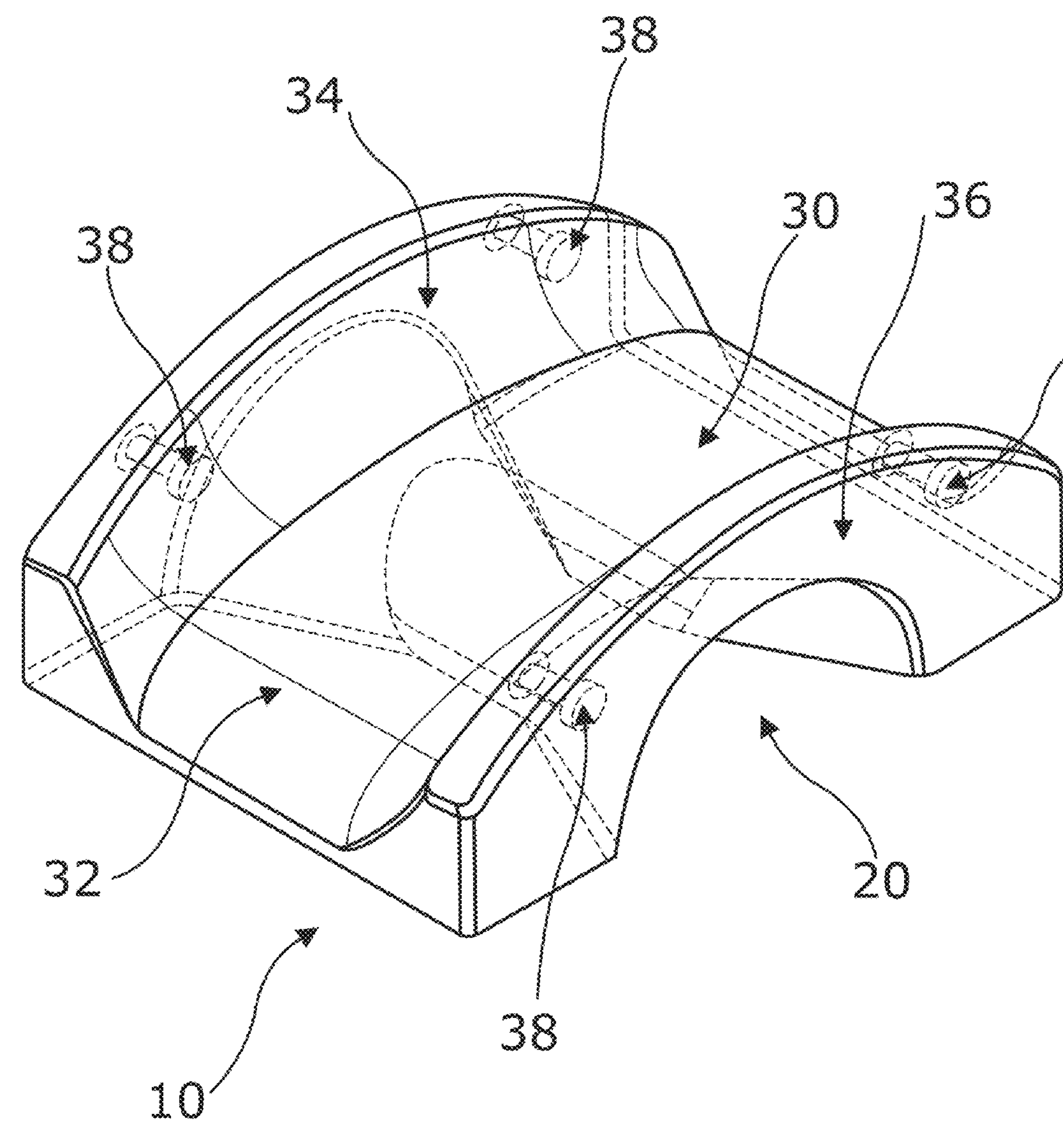
(57) **ABSTRACT**

A crossing bridge for subsea use and a method of installing the crossing bridge is provided. The crossing bridge comprises: a first passageway arranged to enable a first subsea conduit to pass under the crossing bridge; a second passageway arranged to enable a second subsea conduit to pass over the crossing bridge and over the first subsea conduit; one or more chambers for storing seawater when the crossing bridge is submerged in seawater; and one or more valves configured to enable seawater to enter the chamber when the crossing bridge is submerged in seawater.



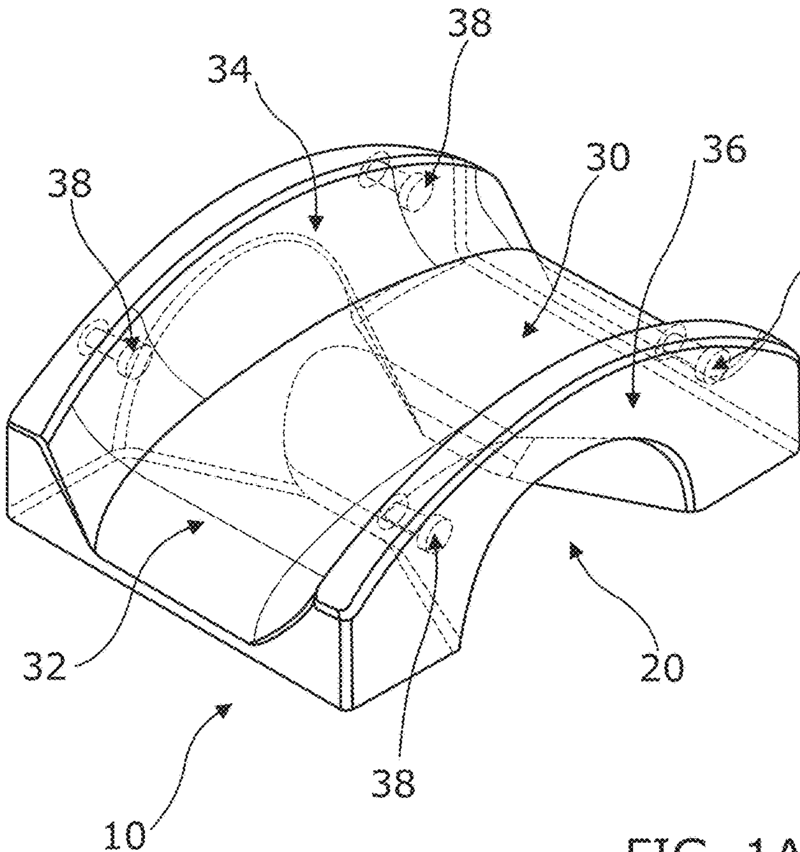


FIG. 1A

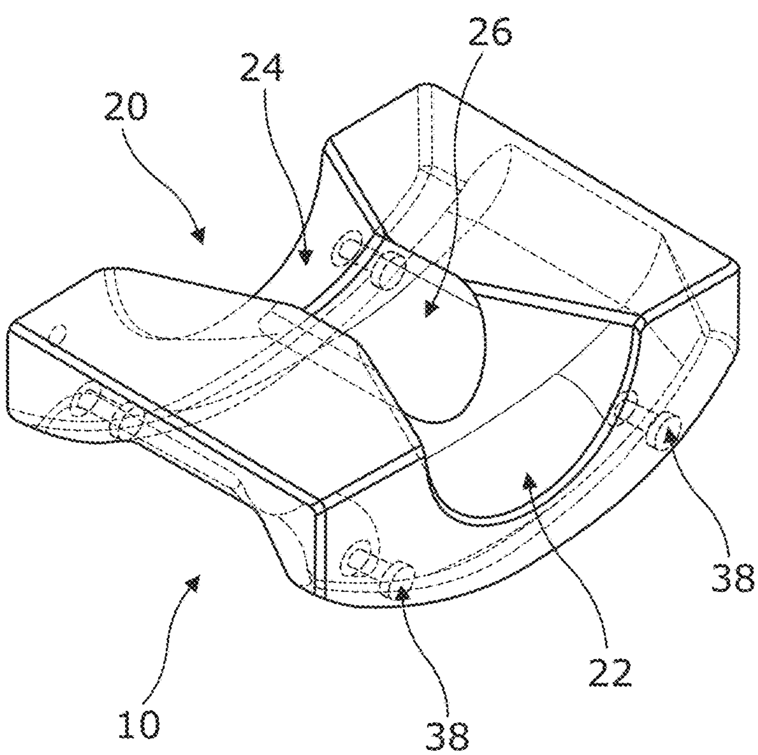


FIG. 1B

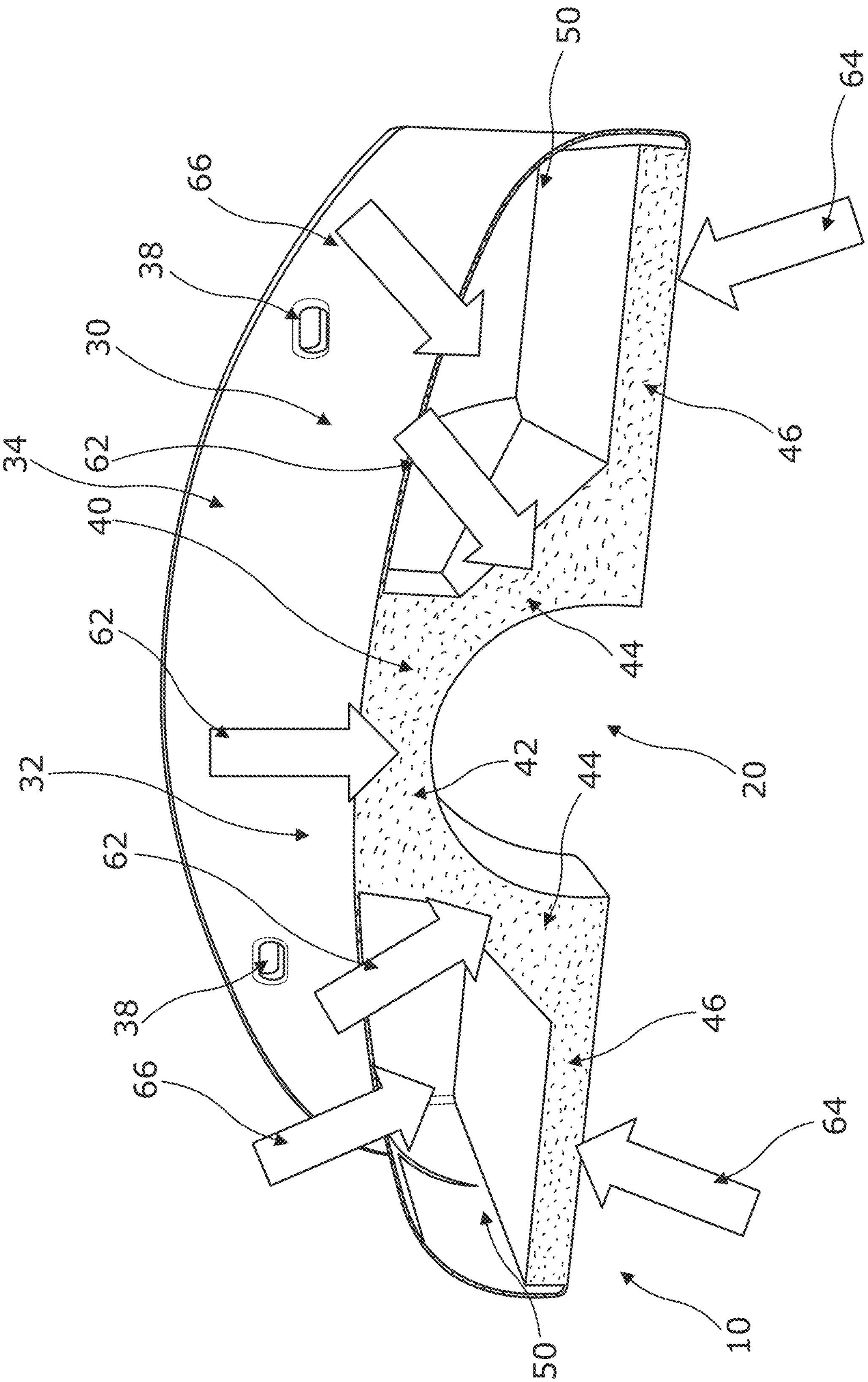


FIG. 2

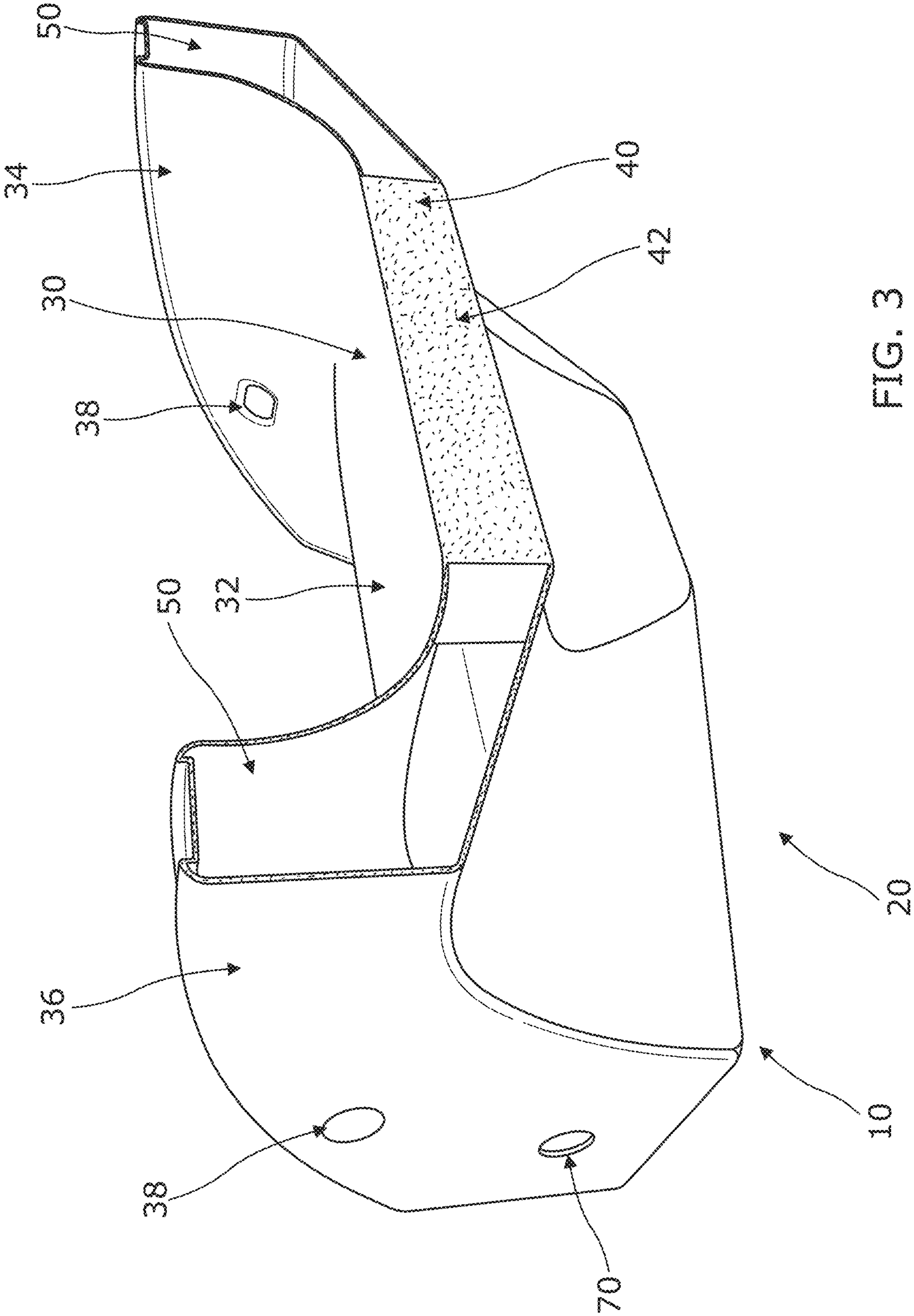


FIG. 3

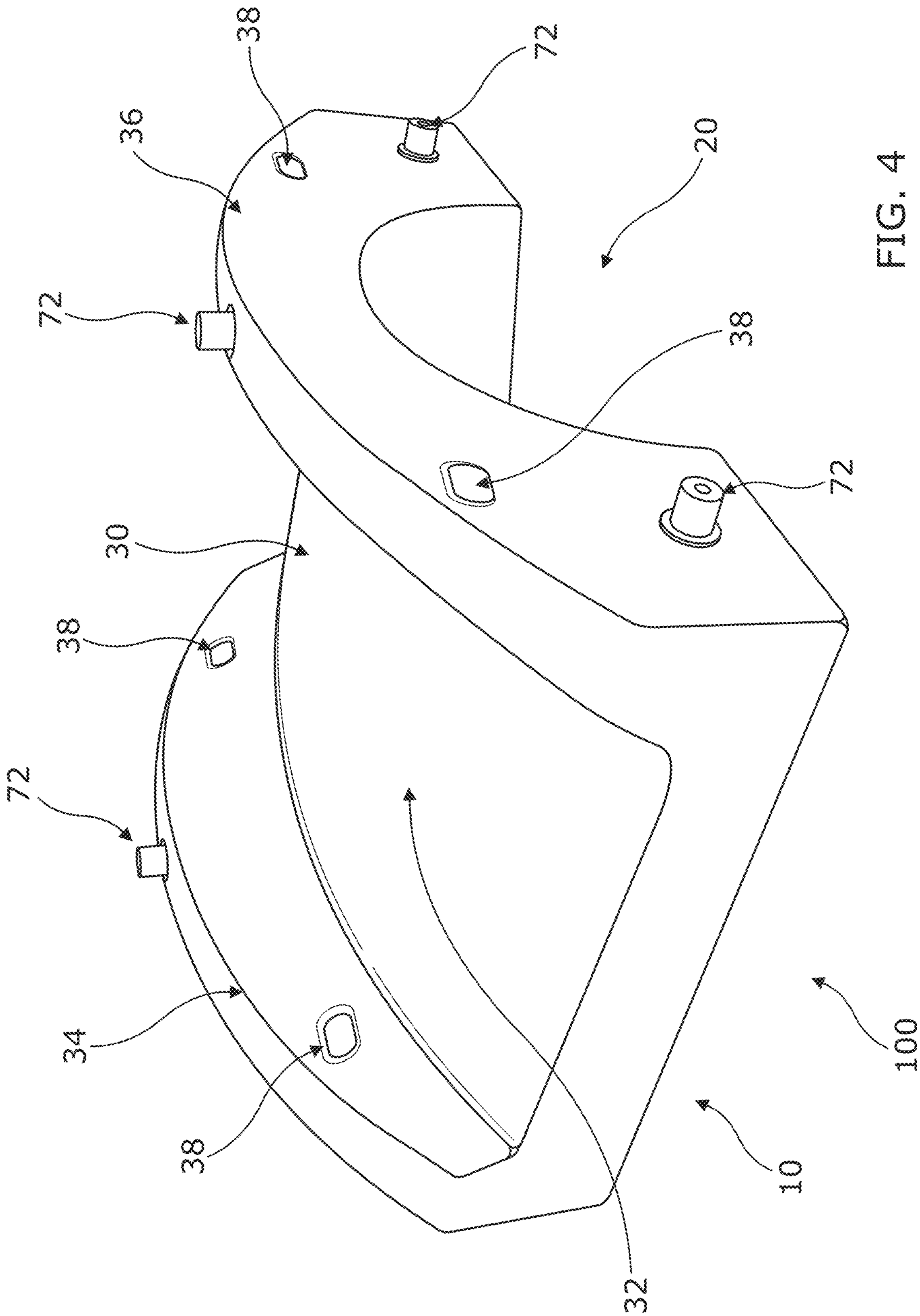


FIG. 4

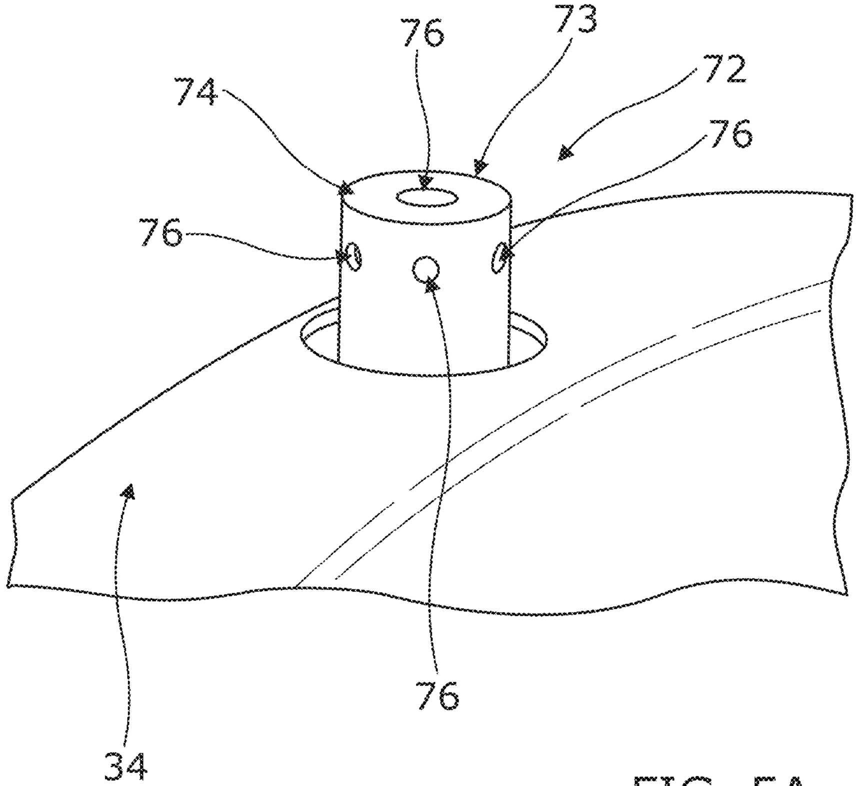


FIG. 5A

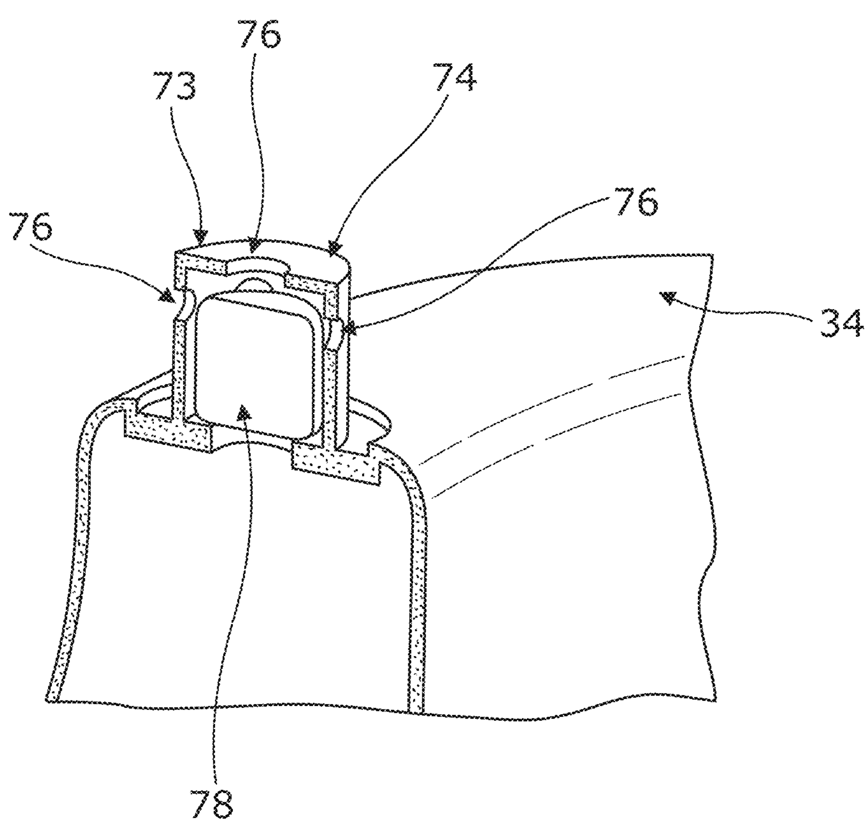


FIG. 5B

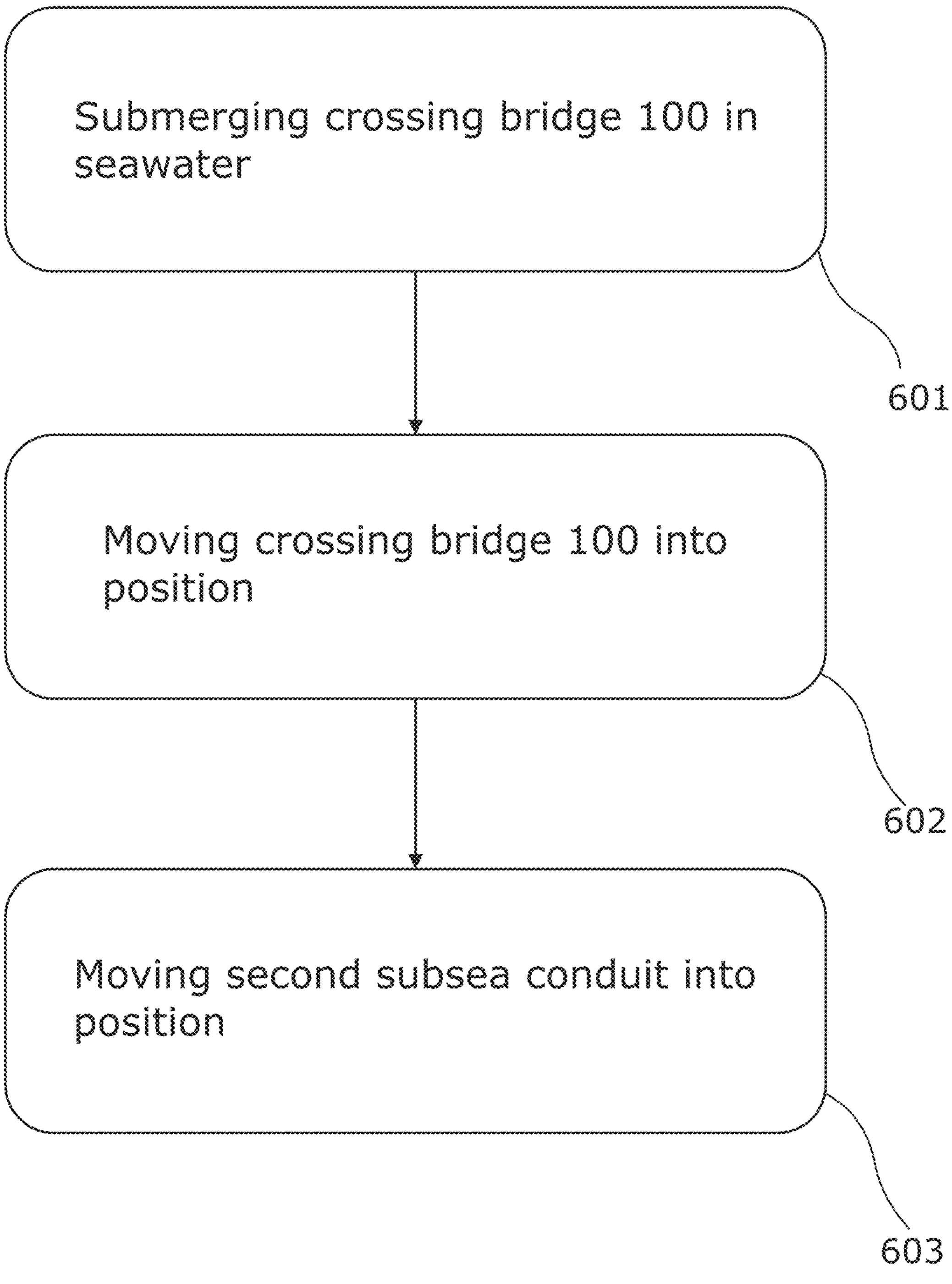

Submerging crossing bridge 100 in seawater
601
Moving crossing bridge 100 into position
602
Moving second subsea conduit into position
603

FIG. 6

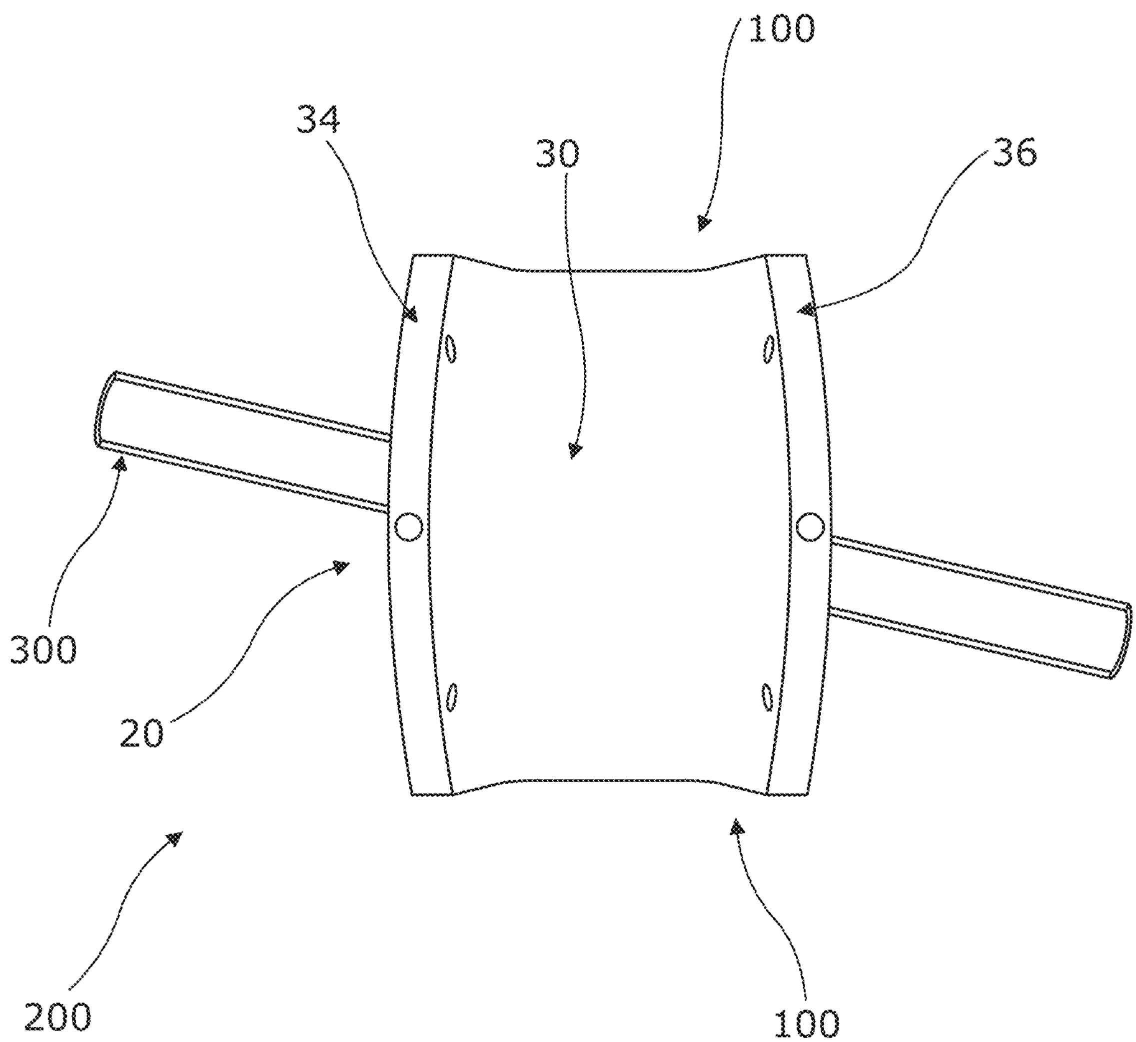


FIG. 7

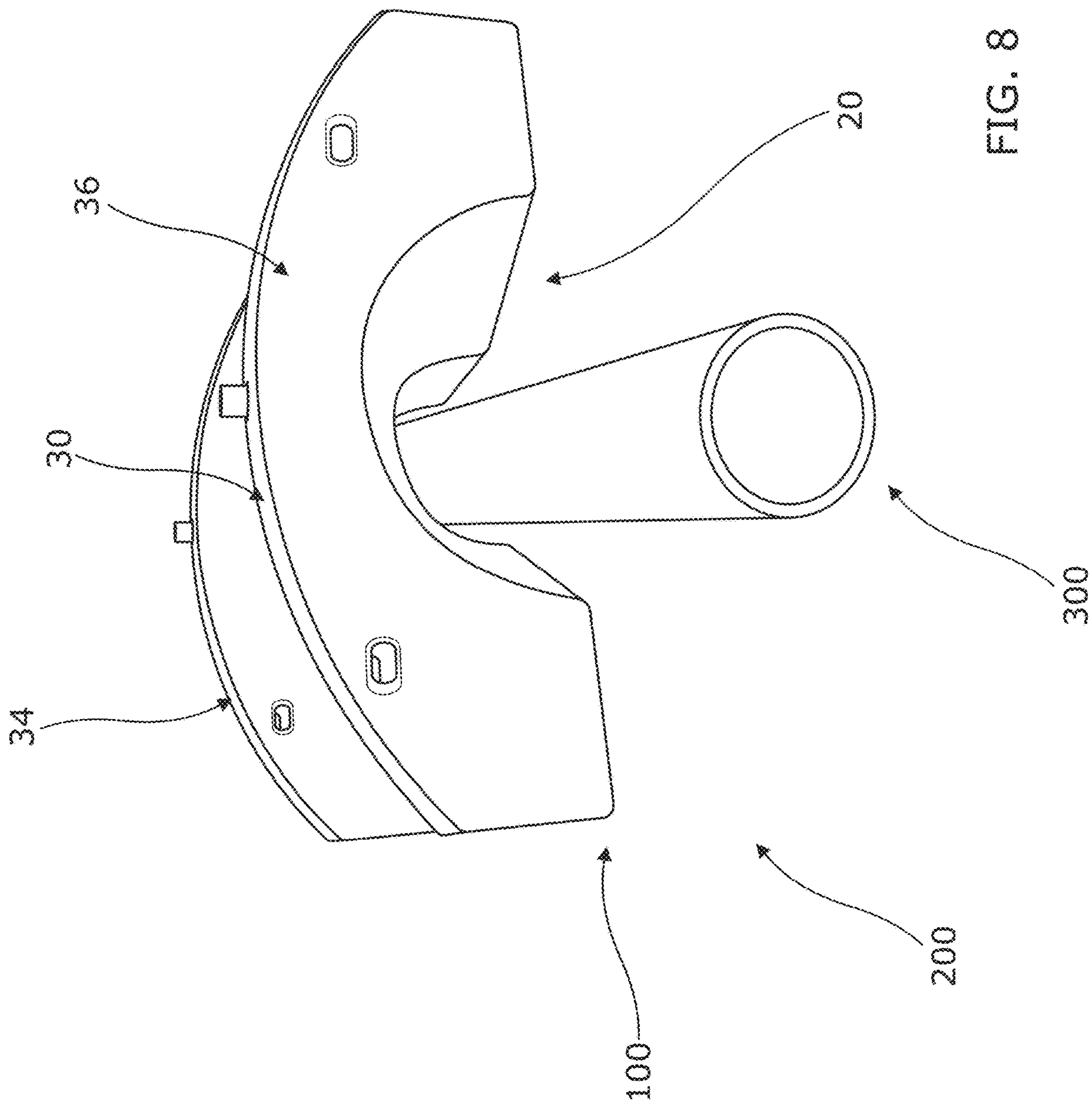


FIG. 8

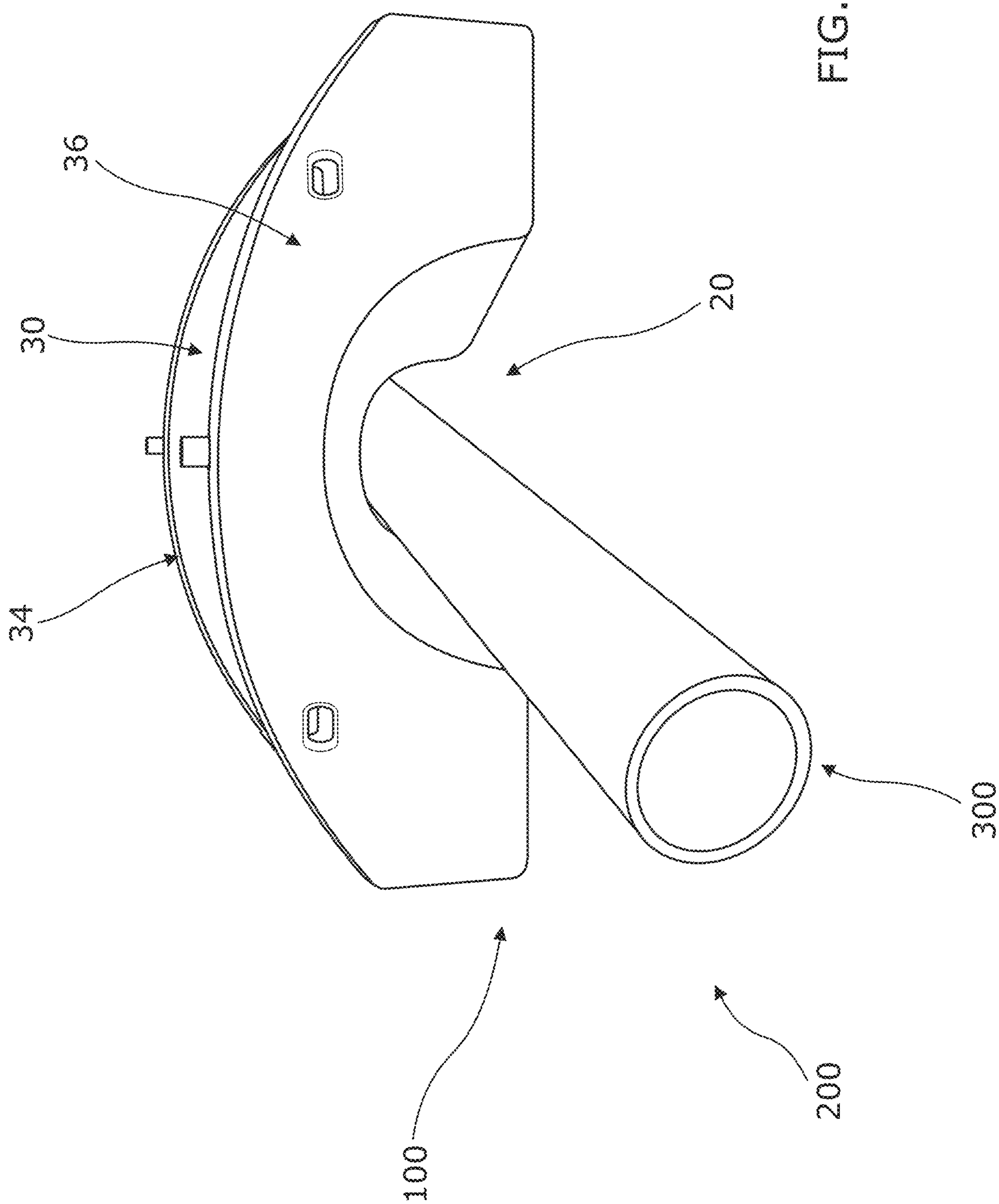


FIG. 9

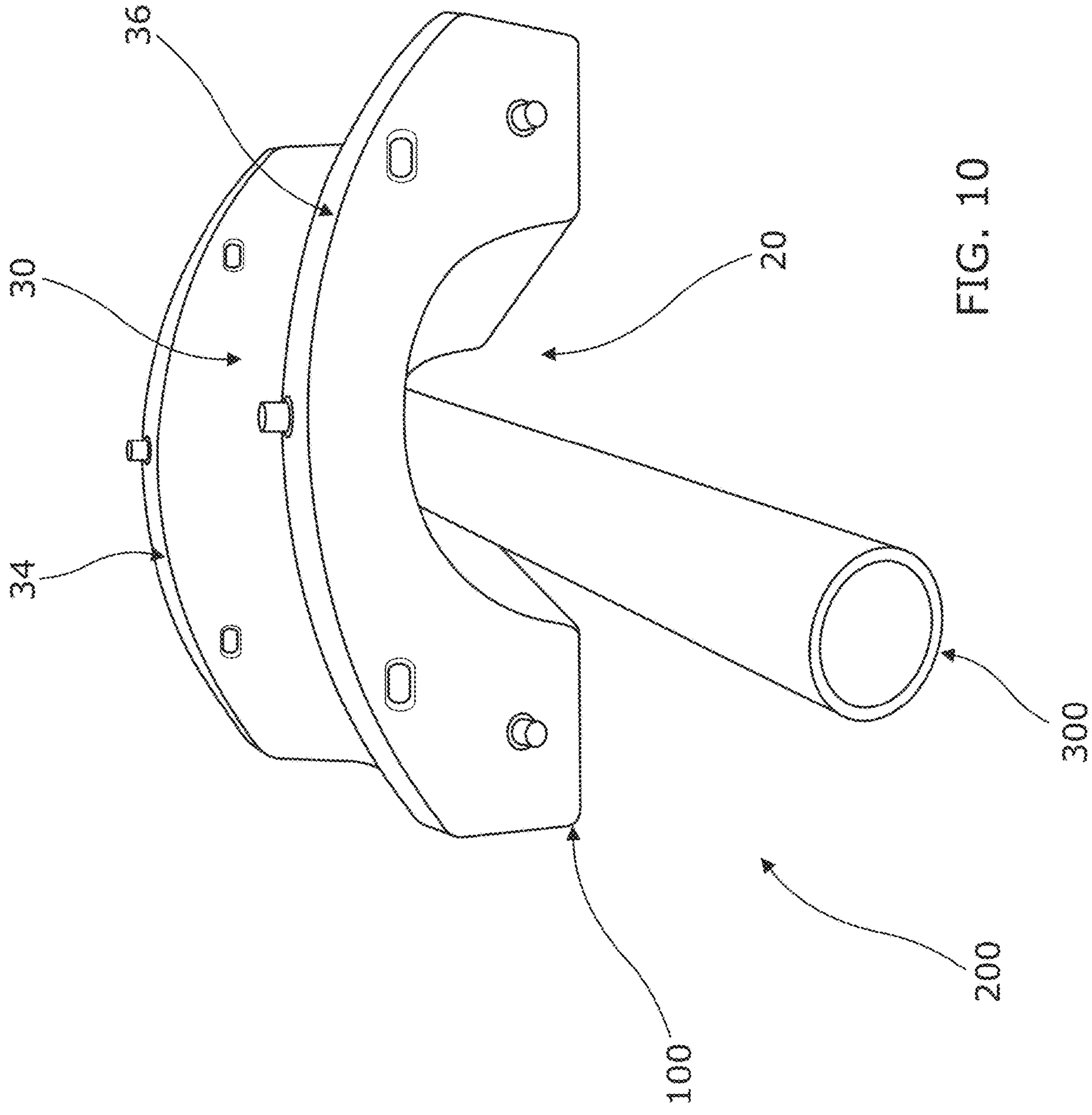


FIG. 10

CROSSING BRIDGE AND METHOD

TECHNOLOGICAL FIELD

[0001] Embodiments of the present disclosure relate to a crossing bridge. Some relate to a crossing bridge for subsea use.

BACKGROUND

[0002] Sometimes it is necessary for subsea conduits such as pipelines or cables to cross one another on the seabed. One conduit may cross another conduit such that the two conduits are perpendicular to one another.

[0003] If the conduits are not properly secured at the crossing, abrasion can occur, which may lead to a fault and a potential hazard. In some circumstances, one conduit may need to be thermally and electrically shielded from the other conduit. This might be the case, for example, if one of the conduits includes a high voltage power cable.

BRIEF SUMMARY

[0004] According to various, but not necessarily all, embodiments there is provided a method, comprising: submerging a crossing bridge in seawater, wherein the crossing bridge comprises a shell having one or more chambers, defining one or more voids in the shell, into which seawater enters when the shell is submerged, and the shell further comprises a first passageway, arranged to enable a first subsea conduit to pass under the shell, and a second passageway, arranged to enable a second subsea conduit to pass over the shell and over the first subsea conduit; moving the crossing bridge into position such that the first subsea conduit passes under the shell and along the first passageway; and moving a second subsea conduit into position such that the second subsea conduit passes along the second passageway and over the first subsea conduit.

[0005] The method may further comprise: after submerging the crossing bridge in seawater, sealing the seawater in the one or more chambers.

[0006] The crossing bridge may further comprise one or more valves which enable the seawater to enter the one or more chambers. Sealing the seawater in the one or more chambers may comprise closing the one or more valves. The one or more valves may comprise one or more hydrophilic valves. Each of the hydrophilic valves may comprise at least one seal that expands in response to absorption of seawater and thereby closes the hydrophilic valve.

[0007] The crossing bridge may be submerged in the seawater by dropping the crossing bridge into the seawater.

[0008] Moving the crossing bridge into position may comprise moving the crossing bridge while the crossing bridge is submerged in seawater. The crossing bridge may be moved by a remotely operated underwater vehicle.

[0009] The method may further comprise: after moving the second subsea conduit into position such that the second subsea conduit passes along the second passageway and over the first subsea conduit, laying rocks over the first subsea conduit, the second subsea conduit and the crossing bridge.

[0010] The first passageway may be a tunnel under the crossing bridge.

[0011] At least a portion of the second passageway may be substantially perpendicular to at least a portion of the first passageway.

[0012] According to various, but not necessarily all, embodiments there is provided a method, comprising: submerging a crossing bridge in water, wherein the crossing bridge comprises a shell having one or more chambers, defining one or more voids in the shell, into which water enters when the shell is submerged, and the shell further comprises a first passageway, arranged to enable a first submerged conduit to pass under the shell, and a second passageway, arranged to enable a second submerged conduit to pass over the shell and over the first submerged conduit; moving the crossing bridge into position such that the first submerged conduit passes under the shell and along the first passageway; and moving a second submerged conduit into position such that the second submerged conduit passes along the second passageway and over the first submerged conduit.

[0013] The method may further comprise: after submerging the crossing bridge in water, sealing the water in the one or more chambers.

[0014] The crossing bridge may comprise one or more valves which enable the water to enter the one or more chambers. Sealing the water in the one or more chambers may comprise closing the one or more valves.

[0015] The one or more valves may comprise one or more hydrophilic valves. Each of the hydrophilic valves may comprise at least one seal that expands in response to absorption of water and thereby closes the hydrophilic valve.

[0016] The crossing bridge may be submerged in the water by dropping the crossing bridge into the water.

[0017] Moving the crossing bridge into position may comprise moving the crossing bridge while the crossing bridge is submerged in water. The crossing bridge may be moved by a remotely operated underwater vehicle.

[0018] The method may further comprise: after moving the second submerged conduit into position such that the second submerged conduit passes along the second passageway and over the first submerged conduit, laying rocks over the first submerged conduit, the second submerged conduit and the crossing bridge.

[0019] The first passageway may be a tunnel under the crossing bridge.

[0020] At least a portion of the second passageway may be substantially perpendicular to at least a portion of the first passageway.

[0021] According to various, but not necessarily all, embodiments there is provided a crossing bridge for subsea use, the crossing bridge comprising: a shell, comprising a first passageway arranged to enable a first subsea conduit to pass under the shell; a second passageway, defined at least in part by a floor of the shell, arranged to enable a second subsea conduit to pass over the shell and over the first subsea conduit; one or more chambers, defining one or more voids in the shell, for storing seawater when the shell is submerged in seawater; and wherein the crossing bridge further comprises: one or more valves configured to enable seawater to enter the one or more chambers when the shell is submerged in seawater.

[0022] The crossing bridge may further comprise at least one structural support configured to support the floor of the shell defining, at least in part, the second passageway.

[0023] In use, at least a portion of the floor of the shell may be located above the first passageway.

[0024] At least a portion of the structural support may be located above the first passageway. The structural support may at least partially define the first passageway. The structural support may be formed, at least in part, from at least one metal.

[0025] At least one chamber of the one or more chambers may be for storing at least one ballast. The crossing bridge may further comprise a plurality of walls arranged, at least in part, to define the second passageway.

[0026] The one or more valves may be located on an exterior surface of the crossing bridge. The one or more valves may be fluidly connected to the one or more chambers. The one or more valves may comprise one or more hydrophilic valves. Each of the hydrophilic valves may comprise at least one seal that is configured to expand in response to absorption of seawater and close the hydrophilic valve.

[0027] The first passageway may be a tunnel under the crossing bridge. At least a portion of the second passageway may be substantially perpendicular to at least a portion of the first passageway. The first passageway may comprise first and second tapered portions that enable the second subsea conduit to be positioned non-perpendicularly relative to the first subsea conduit. The crossing bridge may be an arch bridge.

[0028] According to various, but not necessarily all, embodiments there is provided a crossing bridge for submersion in a liquid, the crossing bridge comprising: a shell, comprising a first passageway arranged to enable a first submerged conduit to pass under the shell; a second passageway, defined at least in part by a floor of the shell, arranged to enable a second submerged conduit to pass over the shell and over the first submerged conduit; one or more chambers, defining one or more voids in the shell, for storing liquid when the shell is submerged in liquid; and wherein the crossing bridge further comprises: one or more valves configured to enable liquid to enter the one or more chambers when the shell is submerged in liquid.

[0029] According to various, but not necessarily all, embodiments there is provided examples as claimed in the appended claims.

BRIEF DESCRIPTION

[0030] Some examples will now be described with reference to the accompanying drawings in which:

[0031] FIG. **1**A illustrates a perspective view of a shell of a crossing bridge;

[0032] FIG. **1**B illustrates a perspective view of an underside of the shell of the crossing bridge;

[0033] FIG. **2** illustrates a cross-sectional view of the shell of the crossing bridge comprising a chamber for storing seawater and a structural support;

[0034] FIG. **3** illustrates another cross-sectional view of the shell of the crossing bridge illustrated in FIG. **2**;

[0035] FIG. **4** illustrates a perspective view of the crossing bridge;

[0036] FIG. **5**A illustrates a perspective view of a hydrophilic valve;

[0037] FIG. **5**B illustrates a perspective view of a cross-section of the hydrophilic valve illustrated in FIG. **5**A;

[0038] FIG. **6** illustrates a flow chart of a method;

[0039] FIG. **7** illustrates a plan view of a schematic of the crossing bridge on the seabed, with a first subsea conduit passing under the crossing bridge on the seabed;

[0040] FIGS. **8**, **9** and **10** illustrate front elevations of schematics of the crossing bridge on the seabed, with a first subsea conduit passing under the crossing bridge on the seabed at different angles.

DETAILED DESCRIPTION

[0041] FIG. **1**A illustrates a perspective view of a shell **10** of a crossing bridge. The shell **10**, and therefore the crossing bridge, comprises a first passageway **20** and a second passageway **30**.

[0042] The first passageway **20** is arranged to enable a first subsea conduit to pass under the crossing bridge. The first subsea conduit may be located on the seabed.

[0043] The first subsea conduit could, for example, be a pipeline for conveying a hydrocarbon mixture such as crude oil or natural gas. Alternatively, the first subsea conduit could be telecommunications cable comprising one or more optical fibres for conveying telecommunications signals, or an electrical cable such as a high voltage cable for conveying electrical power from a wind turbine.

[0044] In this example, the first passageway **20** is a tunnel under the crossing bridge. While the crossing bridge is an arch bridge in the illustration, it need not be in other examples. The shell **10** of the crossing bridge (and therefore the first passageway/tunnel **20**) may have a different shape.

[0045] The second passageway **30** is arranged to enable a second subsea conduit to pass over the crossing bridge and over the first subsea conduit. The second subsea conduit could, for example, be a pipeline for conveying a hydrocarbon mixture such as crude oil or natural gas. Alternatively, the second subsea conduit could be a telecommunications cable comprising one or more optical fibres for conveying telecommunications signals, or an electrical cable such as a high voltage cable for conveying electrical power from a wind turbine. The second subsea conduit may be the same type of conduit as the first subsea conduit, or they may be of different types.

[0046] The second passageway **30** is arranged to extend over the first passageway **20**. At least a portion of the second passageway **30** may be substantially perpendicular to at least a portion of the first passageway **20**.

[0047] The first passageway **30** has a length which represents its longest extent. The second passageway **20** has a length which represents its longest extent. The length of the second passageway **30** is substantially perpendicular to the length of the first passageway **20** in the illustrated example.

[0048] The shell **10** comprises a floor **32** on which the second subsea conduit may be positioned. The second passageway **30** is defined, at least in part, by the floor **32**. In the illustrated example, the floor **32** is curved/arched, although it need not be in every example. In use, at least a portion of the floor **32** is located above the first passageway **20**.

[0049] In the illustrated example, the second passageway **30** is defined, at least in part, by a plurality of (lateral) walls **34**, **36**. Each of the walls **34**, **36** extends upwardly. The floor **32** is located between the lateral walls **34**, **36**. The lateral walls **34**, **36** are arranged to constrain the movement of the second subsea conduit when it is located on the floor **32**. That is, the lateral walls **34**, **36** are arranged to prevent the second subsea conduit from falling off the floor **32** when the second subsea conduit has been placed on it.

[0050] The shell **10** comprises a plurality of connectors **38**. In the illustrated example, each of the connectors **38** is a through hole, but in other examples one, some or all of the connectors **38** may have a different form. The connectors **38** are arranged to enable the shell **10** to be lifted, for example, using a crane comprising a lifting sling. The crane may be used to position the shell **10** above the sea during installation.

[0051] The connectors **38** are located on the lateral walls **34**, **36** in the illustrated example. In other examples they may be positioned elsewhere.

[0052] FIG. **1**B illustrates a perspective view of an underside of the shell **10** of the crossing bridge in which the first passageway **20** is more clearly seen. In this example, the first passageway/tunnel **20** comprises first and second tapered portions **22**, **24** and an intermediate portion **26**. In the first tapered portion **22**, the width of the first passageway **20** tapers inwardly from a first mouth of the first passageway **20** towards the intermediate portion **26** and the second tapered portion **24**. In the second tapered portion **24**, the width of the second passageway **20** tapers inwardly from a second mouth of the first passageway **20** towards the intermediate portion **26** and the first tapered portion **22**.

[0053] In the illustrated example, the intermediate portion **26**, which is located between the first and second tapered portions **22**, **24**, has a substantially constant width, but that might not be the case in other examples. In other examples, the intermediate portion **26** might not be present and first tapered portion **22** might be directly connected to the second tapered portion **24**.

[0054] FIG. **2** illustrates a cross-sectional view of the shell **10** of the crossing bridge.

[0055] The shell **10**, and therefore the crossing bridge, comprises one or more chambers **50** for storing seawater when the shell/crossing bridge is submerged in seawater. Each of the chambers **50** may define a void that can be filled with seawater. In the illustrated example, the shell **10** comprises a single chamber **50**, but in other examples multiple separate chambers **50** may be provided.

[0056] In the example illustrated in FIG. **2**, the crossing bridge comprises at least one structural support **40**. A single structural support **40** is illustrated in FIG. **2**, but multiple structural supports **40** may be provided in other examples. The structural support(s) **40** may be formed, at least in part, from at least one metal. For example, the support(s) **40** may be formed, at least in part, from steel or cast iron.

[0057] In some examples, the structural support **40** may be pre-formed and inserted into the shell **10**. In other examples, one or more separate chambers **50** (i.e., separate from that/those which are used to store seawater) may be provided which can be filled with a ballast material to form the structural support **40**. In still further examples, one or more chambers **50** may be provided which can optionally be used to store seawater, at least one ballast material and/or at least one buoyant material, depending on the application. The ballast material could be sand and/or a metal, such as iron ore.

[0058] In the illustrated example, the structural support **40** defines the first passageway/tunnel **20**. That is, the structural support **40** defines the archway that forms the tunnel **20**.

[0059] The structural support **40** is arranged to support the second passageway **30** and, more specifically, the floor **32** of the second passageway **30**. The structural support **40** comprises one or more base portions **46**, one or more floor supporting portions **42** and one or more interconnecting portions **44**. The structure of the structural support **40** enables at least part of the weight of a second subsea conduit, located on the floor **32**, to be transferred through to the seabed when the crossing bridge is located on the seabed. That is, it enables the weight to be transferred through the floor supporting portion(s) **42**, the interconnecting portion(s) **44** and the base portion(s) **46** through to the seabed. Advantageously, the structure of the crossing bridge enables the weight of the second subsea conduit to be transferred to the seabed without being transferred to the first subsea conduit and without either the crossing bridge of the second subsea conduit being in contact with the first subsea conduit. The separation between the subsea conduits eliminates abrasion, reduces electrical interference and/or reduces heat transfer between the subsea conduits.

[0060] When the shell **10** is submerged in seawater, water enters the chamber(s) **50**. This is described in further detail below. When seawater enters the chamber **50** and is sealed in the chamber **50**, the chamber **50** becomes incompressible. That is, the floor **32** does not deform and thereby reduce the volume of the chamber **50** when a second subsea conduit is placed on the floor **32**.

[0061] The arrows labelled with the reference numeral **62** in FIG. **2** illustrate part of the weight of second subsea conduit being borne by the floor supporting portion **42** and the interconnecting portions **44** of the structural support **40**. The arrows labelled with the reference numeral **64** in FIG. **2** illustrate the support being provided through the base portions **46** of the structural support. The arrows labelled with the reference numeral **66** in FIG. **2** illustrate the support being provided due to the presence of the seawater in the chamber **50**.

[0062] FIG. **3** illustrates another cross-sectional view of the shell **10** of the crossing bridge illustrated in FIG. **2**. This view indicates how the chamber **50** extends through the shell **10**, and in particular through the lateral walls **34**, **36** of the shell **10**.

[0063] FIG. **3** also illustrates that the shell **10** comprises one or more apertures **70** which provide access to the chamber **50**. A valve may be located in each aperture **70**. The valves may be configured to hermetically seal the chamber **50**.

[0064] In examples in which the chamber(s) **50** are used to store ballast material and/or buoyant material, the apertures **70** may be used to insert the ballast material and/or buoyant material into the chamber(s) **50** (e.g., prior to the aperture being plugged or the insertion of a valve in the aperture **70**).

[0065] FIG. **4** illustrates a perspective view of the crossing bridge **100**, which comprises the shell **10** and one or more valves **72**. Each of the one or more valves **72** is located in an aperture **70** in the shell **10**. At least some of the one or more valves **72** may be configured to enable seawater to enter the chamber **50** when the crossing bridge **100** is submerged in seawater. At least some of the one or more valves **72** may be configured to enable air/gas to escape the chamber **50** when the crossing bridge is submerged in seawater. In this regard, each of the one or more valves **72** is fluidly connected to the chamber **50**. It can be seen in FIG. **4** that the valves **72** are located on an exterior surface of the shell **10** and the crossing bridge **100**.

[0066] It may be that at least one of the valves **72** is spaced, in a vertical dimension, from at least one of the other valves **72**. In the illustrated example, multiple valves **72** are spaced in a vertical dimension from multiple other valves **72**. That is, valves **72** on an upper surface of the lateral walls

34, **36** are spaced vertically from valves **72** on a side surface of the lateral walls **34**, **36**. At least a portion of the chamber **50** is located, in the vertical dimension, between the upper valves **72** and the lower valves **72**. In some implementations, a majority or the whole of the chamber **50** is located between the upper valves **72** and the lower valves **72** in the vertical dimension.

[0067] In some examples, one, some or all of the valves **72** may be hydrophilic valves. It will be appreciated, however, that this need not be the case in every example. For instance, in other examples, one, some or all of the valves may be tidal flap valves.

[0068] FIG. **5**A illustrates a perspective view of a hydrophilic valve **72**. FIG. **5**B illustrates a perspective view of a cross-section of the hydrophilic valve illustrated in FIG. **5**A.

[0069] The hydrophilic valve **72** comprises a housing **73** having one or more apertures **76**. The housing **73** defines a chamber **75** in which a seal **78**, formed from a hydrophilic material, is located. The hydrophilic material may be a polymeric material.

[0070] The one or more apertures **76** act as inlets or outlets. For example, the apertures **76** may act as outlets by enabling air/gas located in the chamber **50** to escape the chamber **50** (via the chamber **75** defined by the housing **73**) when the valve **72** is open. Alternatively, the apertures **76** may act as inlets to enable seawater to enter the chamber **75**.

[0071] The seal **78** located in the chamber **75** is configured to expand in response to absorption of seawater and close the hydrophilic valve **72**.

[0072] The seal **78** does not initially fill the volume of the chamber **75**, prior to absorbing seawater. That is, part of the chamber **75** is a void. In response to absorption of the seawater, the seal **78** expands (for example, over the course of days or weeks) and eventually expands to a volume that causes the valve **72** to close, sealing the seawater in the chamber **50**. This prevents seawater from exiting the chamber **50** (e.g., in response to a force being applied to the crossing bridge **100**), rendering the chamber **50** incompressible.

[0073] FIG. **6** illustrates a method of installing the crossing bridge **100** in seawater. In block **601** of FIG. **6**, the crossing bridge **100** is submerged in seawater. The crossing bridge **100** may be lifted by the connectors **38** (e.g., using a crane comprising a lifting sling) and located at an appropriate position above the surface of the sea, such that it can be dropped into position at the location at which it is desired to make a crossing over a first subsea conduit positioned on the seabed. Given the void(s) defined by the chamber(s) **50** of the crossing bridge **100**, the crossing bridge **100** is relatively light to transport.

[0074] When the crossing bridge **100** is dropped into the sea and the crossing bridge **100** is at least partially submerged, seawater begins to enter the chamber(s) **50** via the valves **72** and air begins to exit the chamber(s) **50** via the valves **72**. The seawater ingresses at the lower valves **72** and the air exhausts through the upper valves **72**. The crossing bridge **100** is denser than the seawater, and sinks to seabed. The characteristics of the valves **72** and their location on the crossing bridge **100** will determine the speed at which the crossing bridge **100** sinks to the seabed.

[0075] Seawater will continue to enter the chamber(s) **50** until the chamber(s) **50** are substantially full with seawater. The period of time over which this occurs may depend on the nature of the valves **72** and could be a few hours. As mentioned above, the valves **72** may be configured to hermetically seal the chamber(s) **50**. The period of time between the initial ingress of seawater and the valves **72** being hermetically sealed may depend on the nature of the valves **72**. The hydrophilic valves **72** described above might take a few weeks to hermetically seal the chamber(s) **50**.

[0076] In block **602** in FIG. **6**, the crossing bridge **100** is moved into position such that the first subsea conduit passes along the first passageway **20**, under the crossing bridge **100**. The crossing bridge **100** may be moved, for example, by a remotely operated underwater vehicle. The crossing bridge **100** may be moved into position before or after the chamber (s) **50** are completely full with seawater. The crossing bridge **100** may be moved into position before or after the chamber (s) **50** have been sealed by the valves **72**.

[0077] In block **603** in FIG. **6**, a second subsea conduit is moved into position such that it passes along the second passageway **30**, over the crossing bridge **100** and over the first subsea conduit.

[0078] The crossing point may then be fixed in position by laying rocks over the first subsea conduit, the second subsea conduit and the crossing bridge **100** at the location of the crossing point. Advantageously, the chamber(s) **50** of the crossing bridge **100** are incompressible during the positioning of the second subsea conduit and the laying of the rocks, due to ingress of seawater into the chamber(s) **50** and the hermetic sealing of the valves **72**. In this regard, movement of the second subsea conduit into position and the laying of the rocks are both typically performed after the valves **72** have hermetically sealed the chamber(s) **50**.

[0079] FIG. **7** illustrates a plan view of a schematic of the crossing bridge **100** on the seabed **200**, with a first subsea conduit **300** passing through the first passageway **20**. The first subsea conduit **300** is supported by and is in contact with the seabed **200**.

[0080] FIGS. **8**, **9** and **10** illustrate front elevations of schematics of the crossing bridge **100** on the seabed **200**, with the first subsea conduit **300** passing through the first passageway **20**. If a second subsea conduit were present, it would cross over the bridge **100**. It can be envisaged from these figures that the first and second subsea conduits can be positioned non-perpendicularly relative to each other. This is enabled by the tapered portions **22**, **24** of the first passageway **20**.

[0081] It will be apparent to those skilled in the art that, while embodiments of the invention have been described in relation to use of the crossing bridge on a seabed, they are not limited to such an implementation. For instance, the crossing bridge could be employed in fresh water, or indeed in any liquid.

[0082] The term 'comprise' is used in this document with an inclusive not an exclusive meaning. That is any reference to X comprising Y indicates that X may comprise only one Y or may comprise more than one Y. If it is intended to use 'comprise' with an exclusive meaning then it will be made clear in the context by referring to "comprising only one . . . " or by using "consisting".

[0083] In this description, reference has been made to various examples. The description of features or functions in relation to an example indicates that those features or functions are present in that example. The use of the term 'example' or 'for example' or 'can' or 'may' in the text denotes, whether explicitly stated or not, that such features or functions are present in at least the described example, whether described as an example or not, and that they can be, but are not necessarily, present in some of or all other examples. Thus 'example', 'for example', 'can' or 'may' refers to a particular instance in a class of examples. A property of the instance can be a property of only that instance or a property of the class or a property of a sub-class of the class that includes some but not all of the instances in the class. It is therefore implicitly disclosed that a feature described with reference to one example but not with reference to another example, can where possible be used in that other example as part of a working combination but does not necessarily have to be used in that other example.

[0084] Although examples have been described in the preceding paragraphs with reference to various examples, it should be appreciated that modifications to the examples given can be made without departing from the scope of the claims. For instance, the structural support **40** might not be present in some examples.

[0085] Features described in the preceding description may be used in combinations other than the combinations explicitly described above.

[0086] Although functions have been described with reference to certain features, those functions may be performable by other features whether described or not.

[0087] Although features have been described with reference to certain examples, those features may also be present in other examples whether described or not.

[0088] The term 'a' or 'the' is used in this document with an inclusive not an exclusive meaning. That is any reference to X comprising a/the Y indicates that X may comprise only one Y or may comprise more than one Y unless the context clearly indicates the contrary. If it is intended to use 'a' or 'the' with an exclusive meaning then it will be made clear in the context. In some circumstances the use of 'at least one' or 'one or more' may be used to emphasis an inclusive meaning but the absence of these terms should not be taken to infer any exclusive meaning.

[0089] The presence of a feature (or combination of features) in a claim is a reference to that feature or (combination of features) itself and also to features that achieve substantially the same technical effect (equivalent features). The equivalent features include, for example, features that are variants and achieve substantially the same result in substantially the same way. The equivalent features include, for example, features that perform substantially the same function, in substantially the same way to achieve substantially the same result.

[0090] In this description, reference has been made to various examples using adjectives or adjectival phrases to describe characteristics of the examples. Such a description of a characteristic in relation to an example indicates that the characteristic is present in some examples exactly as described and is present in other examples substantially as described.

[0091] Whilst endeavouring in the foregoing specification to draw attention to those features believed to be of importance it should be understood that the applicant may seek protection via the claims in respect of any patentable feature or combination of features hereinbefore referred to and/or shown in the drawings whether or not emphasis has been placed thereon.

I/we claim:

1.-**11**. (canceled)

12. A method, comprising:

submerging a crossing bridge in water, wherein the crossing bridge comprises a shell having one or more chambers, defining one or more voids in the shell, into which water enters when the shell is submerged, and the shell further comprises a first passageway, arranged to enable a first submerged conduit to pass under the shell, and a second passageway, arranged to enable a second submerged conduit to pass over the shell and over the first submerged conduit;

moving the crossing bridge into position such that the first submerged conduit passes under the shell and along the first passageway; and

moving a second submerged conduit into position such that the second submerged conduit passes along the second passageway and over the first submerged conduit.

13. The method of claim **12**, further comprising: after submerging the crossing bridge in water, sealing the water in the one or more chambers.

14. The method of claim **13**, wherein the crossing bridge comprises one or more valves which enable the water to enter the one or more chambers, and sealing the water in the one or more chambers comprises closing the one or more valves.

15. The method of claim **14**, wherein the one or more valves comprise one or more hydrophilic valves.

16. The method of claim **15**, wherein each of the hydrophilic valves comprises at least one seal that expands in response to absorption of water and thereby closes the hydrophilic valve.

17. The method of claim **12**, wherein the crossing bridge is submerged in the water by dropping the crossing bridge into the water.

18. The method of claim **12**, wherein moving the crossing bridge into position comprises moving the crossing bridge while the crossing bridge is submerged in water.

19. The method of claim **18**, wherein the crossing bridge is moved by a remotely operated underwater vehicle.

20. The method of claim **12**, further comprising: after moving the second submerged conduit into position such that the second submerged conduit passes along the second passageway and over the first submerged conduit, laying rocks over the first submerged conduit, the second submerged conduit and the crossing bridge.

21. The method of claim **12**, wherein the first passageway is a tunnel under the crossing bridge.

22. The method of claim **12**, wherein at least a portion of the second passageway is substantially perpendicular to at least a portion of the first passageway.

23.-**38**. (canceled)

39. A crossing bridge for submersion in a liquid, the crossing bridge comprising:

a shell, comprising:

a first passageway arranged to enable a first submerged conduit to pass under the shell;

a second passageway, defined at least in part by a floor of the shell, arranged to enable a second submerged conduit to pass over the shell and over the first submerged conduit;

one or more chambers, defining one or more voids in the shell, for storing liquid when the shell is submerged in liquid; and

wherein the crossing bridge further comprises:

one or more valves configured to enable liquid to enter the one or more chambers when the shell is submerged in liquid.

40. The crossing bridge of claim **39**, wherein the one or more valves comprise one or more hydrophilic valves.

41. The crossing bridge of claim **40**, wherein each of the hydrophilic valves comprises at least one seal that is configured to expand in response to absorption of seawater and close the hydrophilic valve.

42. (canceled)

43. A crossing bridge for submersion in a liquid, the crossing bridge comprising:

a first passageway arranged to enable a first submerged conduit to pass under the crossing bridge;

a second passageway arranged to enable a second submerged conduit to pass over the crossing bridge and over the first submerged conduit;

one or more chambers for storing liquid when the crossing bridge is submerged in liquid; and

one or more valves configured to enable liquid to enter the one or more chambers when the crossing bridge is submerged in liquid.

44. The method of claim **12**, wherein the water is seawater.

45. The crossing bridge of claim **39**, further comprising at least one structural support configured to support the floor of the shell defining, at least in part, the second passageway.

46. The crossing bridge of claim **39**, wherein, in use, at least a portion of the floor of the shell is located above the first passageway.

47. The crossing bridge of claim **43**, wherein the one or more valves comprise one or more hydrophilic valves.

48. The crossing bridge of claim **47**, wherein each of the hydrophilic valves comprises at least one seal that is configured to expand in response to absorption of seawater and close the hydrophilic valve.

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